

Fermionic Matrix Models, Fortuity, and the Bootstrap

Abstract. We study a three-parameter family of purely fermionic gauge quantum mechanics, p flavours of adjoint matter for a rank- r algebra with a nilpotent supercharge of odd degree q , using two methods whose strengths are complementary. Exact cohomology answers questions about which states are BPS: because Q acts on the Fock space as multiplication by a fixed q -form, the maximal irrep factorises over Cartan sites and its BPS window has width $W = r(w_s - 1) + 1$ in closed form, where the slot width w_s takes only the values $q - 1$, q and $q + 1$ and is controlled by the symmetry type of one bilinear form. A combinatorial criterion based on critical sets gives a lower bound on W at every weight, and exact integer linear algebra over a weight decomposition, followed by Kostant inversion, gives BPS multiplicities irrep by irrep for N up to 5. Since $\mathcal{N} = 2$ SYK is the $N = 1$ member of the same family, R-charge concentration becomes a statement about rank: it holds for $r \leq q - 1$ and fails otherwise. None of this reaches the density of states, which is what distinguishes a chaotic model with a super-Schwarzian tail from a solvable one, so for that we use a sector-resolved bootstrap. A semidefinite programme over linear functionals, formulated exactly at finite N in terms of trace words, gives rigorous lower bounds on the sector ground energies, reproduces them exactly in every gapped sector at $N = 2$, and yields the closed form $E_0(k; N) = 16N^3 - (15 + 38k)N$ for the non-interacting sectors $k \leq N - 1$. Adding the BPS conditions turns the same programme into a certificate of absence, and resolving it by gauge Casimir makes the two methods meet: at $N = 3$ the exclusion margin in the maximal-Casimir irrep decays monotonically through $k = 8, 9, 10, 11$ and vanishes precisely at $k = 12$, which is where the exact cohomology places the edge of the window. We close with the single-trace forms of H , Q and $\hat{C}_2$, and with the demonstration that the three-flavour Hamiltonian is not a function of Casimirs, which is what forces the bootstrap to be sector-resolved in the first place.

1 Introduction

1.1 The models

Let $\mathfrak{g}$ be a reductive Lie algebra of rank r and let Ψ^a , $a = 1, \dots, p$, be complex fermions valued in $\mathfrak{g}$. For $\mathfrak{g} = \mathfrak{gl}(N)$ we write Ψ_{ij}^a with $\{\Psi_{ij}^a, \bar{\Psi}_{kl}^b\} = \delta^{ab}\delta_{il}\delta_{jk}$, the gauge group acting by conjugation. The Fock space is the exterior algebra

$$\mathcal{H} = \Lambda^\bullet V, \quad V = \mathbb{C}^p \otimes \mathfrak{g}, \quad n \equiv \dim V = p \dim \mathfrak{g}, \quad (1)$$

graded by the R-charge $k = N_\Psi = \text{Tr} \Psi^a \bar{\Psi}^a$. All dynamics comes from one supercharge,

$$Q = \sum_{a_1, \dots, a_q=1}^p C_{a_1 \dots a_q} \text{Tr}[\Psi^{a_1} \dots \Psi^{a_q}], \quad H = \{Q, \bar{Q}\} \geq 0, \quad [N_\Psi, Q] = qQ. \quad (2)$$

The flavour sum runs over all tuples. Moving an odd matrix around the trace costs $(-1)^{q-1} = +1$, so the trace is cyclically but not totally antisymmetric in the flavours and only the cyclic-invariant part of C contributes; terms with repeated flavours such as $\text{Tr}[\Psi^1 \Psi^1 \Psi^2]$ are non-zero. Only odd q is admissible, for two reasons: a cyclic shift gives $\text{Tr}[\Psi^q] = (-1)^{q-1} \text{Tr}[\Psi^q]$, so the trace vanishes identically for even q ; and $Q^2 = 0$ requires the exchange of the two q -tuples in $C \otimes C$, a permutation of $2q$ fermions built from q^2 transpositions, to produce a relative sign.

Every Ψ is a creation operator, so Q contains no annihilators and acts on $\Lambda^\bullet V$ as plain left multiplication,

$$Q = \omega \wedge -, \quad \omega = \sum C_{a_1 \dots a_q} \text{Tr}[\Psi^{a_1} \dots \Psi^{a_q}] \in \Lambda^q V, \quad H^\bullet = \text{Ann}(\omega)/(\omega). \quad (3)$$

Nilpotency is then automatic for odd q , since $\omega \wedge \omega = -\omega \wedge \omega$, and survives any change of couplings or of index structure. The whole BPS problem is the cohomology of multiplication by one specific element of $\Lambda^q V$.

1.2 SYK is the member at $N = 1$

The family is usually presented as a matrix generalisation of SYK, but SYK is already inside it. At $N = 1$ the matrices are 1×1 , the trace is the identity map, and (2) becomes $Q = \sum_{a_1 < \dots < a_q} C_{a_1 \dots a_q} \psi^{a_1} \dots \psi^{a_q}$ with C now totally antisymmetric, which is $\mathcal{N} = 2$ SYK with p complex fermions. The gauge algebra is $u(1)$, whose adjoint action is trivial, so nothing constrains C ; this is why the SYK coupling is free while the matrix coupling is not. Randomness plays no role in what follows: what a Gaussian ensemble supplies is genericity, and genericity is optimal here because matrix ranks are lower semicontinuous in ω , so specialising the couplings can only enlarge

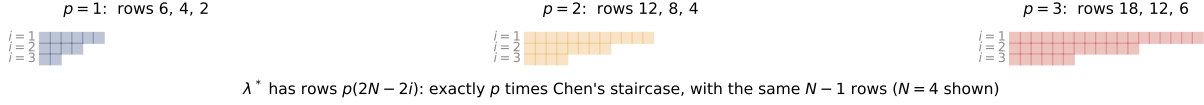


Figure 1: The highest weight occurring in the Fock space, shifted to a partition. Adding flavours stretches the rows without producing new ones, so λ^* has the same $N - 1$ rows for every p .

the cohomology. We checked the identification rather than assuming it, running the full matrix machinery at $N = 1$ against a bare generic q -form and finding identical Betti numbers for every p from 4 to 12.

What varies along the family is not the coupling tensor but the interaction hypergraph. Calling a *triangle* the support of a non-zero term of ω , SYK has all $\binom{p}{q}$ of them, whereas at general N only the index-loop triangles survive out of the $\binom{pN^2}{q}$ available, a fraction of order $6/N^3$ at $q = 3$. The parameter N dials the hypergraph from complete to sparse.

1.3 What each method is for

The Cartan torus acts on Ψ_{ij} by $u_i u_j^{-1}$, so that mode carries weight $e_i - e_j$ and a state's weight is the row sums minus the column sums of its occupation matrix. Since Q is a gauge singlet it preserves the weight, and since $[N_\Psi, Q] = qQ$ it raises k by q , so the cohomology splits into complexes labelled by a $U(N)$ irrep λ , a flavour charge, and $c = k \bmod q$. R-charge concentration, in the sense of Chang, Chen, Sia and Yang [2], says that in each complex of macroscopic index all BPS states sit at one degree. Writing W for the number of consecutive R-charges carrying BPS states, concentration requires $W \leq q$, since a window wider than the grading period must put two occupied degrees in one residue class.

Section 2 answers that question with exact methods. Cohomology is the right tool there because the BPS condition is cohomological, the complexes are finite-dimensional, and the answer is an integer that exact linear algebra can certify. What cohomology cannot see is energy. It says which states have $E = 0$ and says nothing whatever about the states just above, and it is the density of states near the BPS edge that distinguishes a chaotic model with a near-BPS super-Schwarzian tail [7] from a solvable one with a gap fixed by a Casimir. Section 3 therefore turns to a sector-resolved bootstrap, which produces rigorous lower bounds on E_0 in each R-charge sector and, when the gauge Casimir is added as a linear constraint, in each irrep. The two methods then meet: the bootstrap's BPS-exclusion margin, computed with no knowledge of the cohomology, vanishes exactly where the cohomology places the edge of the window.

2 BPS spectra and the R-charge window

2.1 Irreps and the maximal weight

Since each mode has weight $e_i - e_j$ of total zero, only weights with $\sum_i \lambda_i = 0$ occur, and the occurring dominant weights are exactly those dominated by the maximal one. The maximal weight itself is found greedily, filling row 1 and emptying column 1, then row 2, and so on:

$$\lambda_i^* = p(N + 1 - 2i), \quad C_2^{\max} = \frac{1}{2}p(p+1) \sum_i (N + 1 - 2i)^2 = \frac{p(p+1)N(N^2-1)}{6}, \quad (4)$$

using $\lambda_i^* + N + 1 - 2i = (p+1)(N + 1 - 2i)$ and $\sum_i (N + 1 - 2i)^2 = N(N^2 - 1)/3$. At $p = 1$ this is $N(N^2 - 1)/3$, which is Chen's $C_2(r_*)$, so (4) is its multi-flavour generalisation; at $p = 3$ it gives $2N(N^2 - 1) = 48, 120, 240$ for $N = 3, 4, 5$. Shifting λ^* to a partition gives row lengths $p(2N - 2i)$, exactly p times Chen's staircase with the same $N - 1$ non-empty rows (Fig. 1). Counting the occurring irreps gives 25 at $N = 3$, 213 at $N = 4$ and 2131 at $N = 5$; the box bound $|\lambda_i| \leq p(N - 1)$ is necessary but not sufficient, and dominance is the exact criterion.

2.2 Index loops and the width at the maximal weight

At λ^* the off-diagonal sector is rigid, so the weight space is an exterior algebra on the zero-weight modes, $(\Lambda V)_{\lambda^*} \cong \Lambda^\bullet(\mathcal{Z})$ with $\mathcal{Z} = \mathfrak{h} \otimes \mathbb{C}^p$ of dimension pr . Which terms of Q survive is best seen graphically. Read a mode Ψ_{ij}^a as a directed edge $i \rightarrow j$ of colour a . A term is a trace, so the q modes it creates form a closed walk, and Pauli exclusion lets it act only if every edge of that walk is empty. At the maximal weight the empty modes are exactly those with $i > j$, together with free diagonal flavours, so every step of a creatable walk satisfies

(a) maximal weight freezes the off-diagonal

	$j=1$	$j=2$	$j=3$
$i=1$	free	p	p
$i=2$	0	free	p
$i=3$	0	0	free

occupation n_{ij} at λ^*

(b) only self-loops can close a cycle

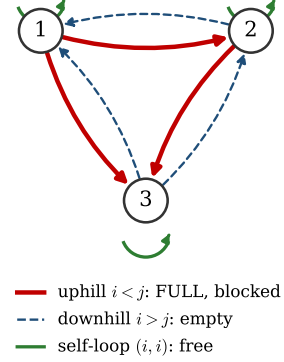


Figure 2: (a) At the maximal weight the occupation is forced: every mode with $i < j$ is full, every mode with $i > j$ is empty, and only the diagonal is free. (b) Reading modes as coloured directed edges, Q creates a closed walk $i \rightarrow j \rightarrow k \rightarrow i$ and needs all its edges empty. Uphill edges are occupied, so every creatable step is non-increasing; a cycle of non-increasing steps must be constant, leaving only self-loops.

N	window (N_Ψ)	W	BPS multiplets h^k
2	5, 6, 7	3	9, 18, 9
3	12, ..., 15	4	27, 81, 81, 27
4	22, ..., 26	5	81, 324, 486, 324, 81
5	35, ..., 40	6	243, 1215, 2430, 2430, 1215, 243

Table 1: Exact BPS multiplets of the maximal irrep at $p = q = 3$, as predicted by (5): $3^N \binom{N}{j}$ over $N + 1$ consecutive charges centred on $pN^2/2$, totalling 6^N . The factor $(1+t)^N$ is the same one that appears in Chen's single-matrix Z_{BPS} .

$i_t \geq i_{t+1}$: the walk may descend or stay level but never climb, because all uphill edges are occupied. Closing it forces $i_1 = \dots = i_q$. Only self-loops survive, meaning q diagonal modes of distinct flavours at a single site, which we verified exhaustively at $N = 3, p = 3$, where all 192 non-zero actions of Q on the maximal-weight space are of that form.

Because every surviving term sits at one site, $Q = \sum_{i=1}^r Q_i$ with $Q_i = \omega_{\text{sl}} \wedge$ on $\Lambda^\bullet \mathbb{C}^p$ and $\omega_{\text{sl}} = \text{Alt}(C)$, the Q_i act on disjoint tensor factors of $\Lambda(\mathcal{Z}) = (\Lambda^\bullet \mathbb{C}^p)^{\otimes r}$, so Künneth applies and Poincaré polynomials multiply,

$$Z_{\text{max}}(t) = t^{k_0} (z_{\text{slot}}(t))^r, \quad z_{\text{slot}}(t) = \sum_j \dim H^j(\Lambda^\bullet \mathbb{C}^p, \omega_{\text{sl}} \wedge) t^j, \quad (5)$$

with $k_0 = p|\Delta^+|$ the number of frozen off-diagonal modes. Multiplying r copies adds degree spans, so if z_{slot} is supported on w_s degrees then

$$W = r(w_s - 1) + 1. \quad (6)$$

It is worth running this once. At $p = q = 3$ and $N = 3$ the slot algebra has dimensions 1, 3, 3, 1 and $\text{Alt}(C)$ spans the one-dimensional $\Lambda^3 \mathbb{C}^3$, so $\omega_{\text{sl}} \wedge$ is non-zero only as $\Lambda^0 \rightarrow \Lambda^3$; hence $H^0 = 0$, $H^1 = H^2 = 3$, $H^3 = 0$, and $z_{\text{slot}} = 3t(1+t)$. Cubing and restoring the prefactor gives $Z_{\text{max}} = 27t^{12} + 81t^{13} + 81t^{14} + 27t^{15}$, in exact agreement with computation.

2.3 What this says about SYK

Setting $r = 1$ in (6) gives $W = w_s$, and $r = 1$ is the $N = 1$ member, so the rank law already contains SYK. This is not a coincidence of notation: z_{slot} is by definition the cohomology of a generic q -form on $\mathbb{C}^p$, which is the SYK computation. One formula therefore covers both ends of the family, and concentration becomes the condition

$$r(w_s - 1) \leq q - 1. \quad (7)$$

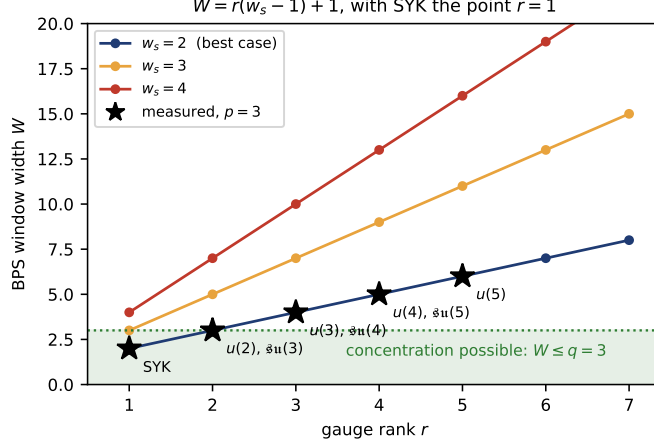


Figure 3: The rank law with the measured models on it. SYK is not a separate case but the point $r = 1$; concentration survives exactly as far as rank two.

Every model we have measured has $W = r + 1$, including SYK (Fig. 3). That $u(2)$ and $\mathfrak{su}(3)$, different algebras of different N and different dimension, give the same W is a non-trivial check that rank rather than N or $\dim \mathfrak{g}$ is the governing variable.

2.4 The slot width takes three values

Everything reduces to w_s . Grade $\Lambda^\bullet \mathbb{C}^p$ by $j \bmod q$ and let

$$E_c = \sum_{j \equiv c \pmod{q}} (-1)^j \binom{p}{j} = \frac{1}{q} \sum_{m=0}^{q-1} \zeta^{-mc} (1 - \zeta^m)^p, \quad \zeta = e^{2\pi i/q}, \quad (8)$$

be the Witten index of class c , a topological invariant independent of the coupling. Since $\sum_c E_c = (1 - 1)^p = 0$, exactly one non-zero E_c is impossible, and since no factor $1 - \zeta^m$ vanishes they cannot all be zero, so at least two classes carry cohomology. This gives $w_s \geq 2$ and hence $W \geq r + 1$: concentration already requires $r \leq q - 1$ whatever the flavour content.

Computation says more. For generic couplings every class with $E_c \neq 0$ carries cohomology in a single degree, of dimension exactly $|E_c|$, with no exception in twenty-seven cases checked class by class at $q = 3, 5, 7$. At most one class ever has $E_c = 0$, and it is either empty or carries two degrees, so

$$w_s = \begin{cases} q - 1, & \text{the zero-index class is empty,} \\ q, & \text{no class has vanishing index,} \\ q + 1, & \text{the zero-index class is occupied.} \end{cases} \quad (9)$$

Which case occurs is settled by a connecting homomorphism. Adjoining one flavour writes $\omega_{p+1} = \alpha + e \wedge \beta$ and, using that β has even degree while α has odd degree,

$$\omega_{p+1} \wedge (x + e y) = \alpha x + e (\beta x - \alpha y), \quad (10)$$

so the new complex is two copies of the old one glued by β . Since $q(q - 1)$ is even, α and β commute and β descends to $\delta : H_p^j \rightarrow H_p^{j+q-1}$, the connecting map of the resulting long exact sequence. It raises degree by $q - 1$, hence sends class c to class $c - 1$, and can be non-zero only between two occupied degrees exactly $q - 1$ apart. When $w_s = q$ there is precisely one such map, and particle-hole conjugation puts its source and target at j and $p - j$ with $j = \frac{1}{2}(p - q + 1)$, so they are Poincaré dual and δ is a bilinear form, $\langle a, b \rangle_\delta = [\beta \wedge a \wedge b]$, with $\langle b, a \rangle_\delta = (-1)^j \langle a, b \rangle_\delta$. A generic symmetric form is non-degenerate, so w_s drops to $q - 1$; an antisymmetric form on an odd-dimensional space cannot be, so w_s rises to $q + 1$. Hence w_s cycles through $q - 1, q, q + 1, q$ with period four in p , turning at $p \equiv q + 1 \pmod{4}$. Building δ explicitly over $\mathbb{F}_p$ confirms this in all nine available cases.

Two consequences follow. Since $w_s \geq q - 1$, the best available rank law is $W = r(q - 2) + 1$, so (7) permits rank 2 at $q = 3$ and only rank 1, that is SYK itself, for every $q \geq 5$: the cubic supercharge is the only degree that tolerates a gauge group. And since $w_s \geq 2$ always, a bounded window with $r \rightarrow \infty$ is impossible, so concentration is a property of a family whose parameter, for a matrix model, must be the rank.

a state S is CRITICAL iff every triangle straddles (filled = occupied mode, open = empty)

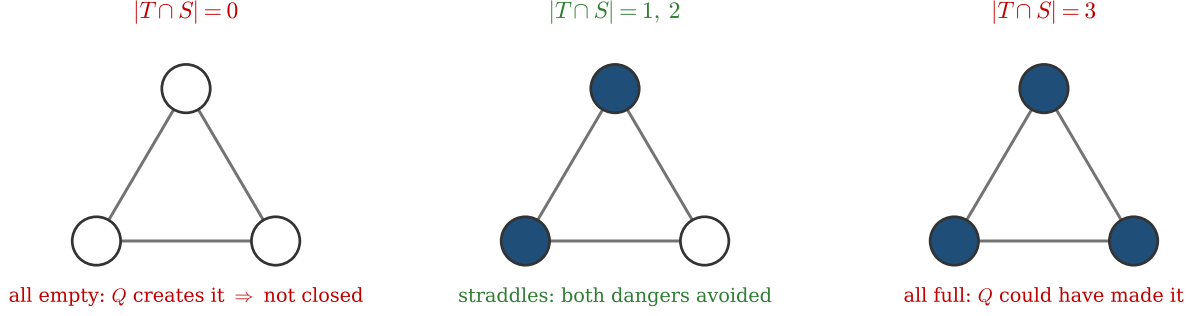


Figure 4: The three cases. If a triangle is entirely empty, Q can create it and S is not closed. If it is entirely occupied, S could have been produced by Q and the non-exactness argument fails. Only straddling triangles avoid both.

p	τ (closed form)	τ (exact integer program)	bound (12)	measured W
1	$N(N-1)/2$	3, 6, 10, 15	$N+1$	$N+1$ (tight)
2	$pN(N+1)/2 - 2N$	6, 12, 20	$2N+1$	$2N+1$ (tight)
3	$pN(N+1)/2 - 2N$	5, 12, 22, 35	$N+1$	$N+1$ (tight)
4	$pN(N+1)/2 - 2N$	18	1	$2N+1$ (valid, loose)

Table 2: The transversal number and the resulting bound for single-trace $u(N)$. The closed form agrees with an exact integer program in all twelve cases tested (p from 1 to 4, N from 2 to 6). The bound is attained for $p \leq 3$; at $p = 4$ the family A_s is no longer optimal.

2.5 A lower bound at every weight

Equation (6) is derived at the maximal weight. A weaker statement holds everywhere, by an elementary argument that also exposes the mechanism. Say that a set of modes S is *critical* if every triangle meets it partially, $0 < |T \cap S| < q$; equivalently, neither S nor its complement contains a triangle. The two inequalities do different jobs (Fig. 4). The upper one makes m_S closed: for $\omega_T \wedge m_S$ to be non-zero all q modes of T must be absent from S , which criticality excludes, so every term is Pauli-blocked. The lower one makes m_S non-exact: since Q only ever adds a triangle, the coefficient of m_S in $\omega \wedge y$ is $\sum_{T \subseteq S} \pm c_T y_{S \setminus T}$, and criticality gives $T \not\subseteq S$ for every triangle, so that sum is empty for every y . Each critical set therefore gives a non-zero class in $H^{|S|}$, and distinct ones give independent classes.

Letting α be the largest triangle-free set and $\tau = n - \alpha$ the transversal number, S is critical exactly when both S and S^c are hitting sets, so critical sets live in $\tau \leq |S| \leq n - \tau$ and

$$W \geq n - 2\tau + 1 = 2\alpha - n + 1. \quad (11)$$

Gauge invariance supplies a large triangle-free set for free. Each term of Q is a singlet, so the weights of its q modes sum to zero, while inside $\mathfrak{n}^+ \otimes \mathbb{C}^p$ every mode carries a positive root and a non-empty sum of positive roots is never zero. No triangle fits in the positive-root subspace, for any reductive $\mathfrak{g}$ and any invariant coupling, so $\alpha \geq \frac{1}{2}(n - pr)$ and the root contribution cancels in (11), leaving $W \geq 2\alpha_{\text{cart}} - pr + 1$ in terms of the Cartan sector alone. For single-trace $u(N)$ a trace of q diagonal matrices localises to one site, so $\alpha_{\text{cart}} = \min(p, q-1)N$ and

$$W \geq N(2\min(p, q-1) - p) + 1, \quad (12)$$

which at $p = 1$ reads $W \geq N + 1$, recovering Chen's single-matrix window combinatorially with no Casimir argument. The explicit critical sets are $A_s = \{\text{all } p \text{ flavours of } E_{ij}, i < j\} \cup \{s_i \text{ flavours of } E_{ii}\}$ with $\max(0, p - q + 1) \leq s_i \leq \min(p, q - 1)$: the strictly upper part admits no closed loop and each site is one flavour short of a self-loop, and the complement is triangle-free for the same reason.

The immediate consequence is a no-go: for $p \leq q - 1$ we get $W \geq pN + 1$, so concentration fails for every $N > (q - 1)/p$, and at $p = q = 3$ this reads $W \geq N + 1 > q$ for all $N \geq 3$. Table 2 shows the bound against what is measured.

N	irreps computed	Casimir range	window	total BPS states	index-invisible
2	all 4	12 down to 0	5...7	972	0%
3	13 of 25	48 down to 24	12...15	1 914 064	10.1%
4	13 of 213	120 down to 107	22...26	3 008 825 568	19.0%
5	maximal only	240	35...40	$6^5 = 7776$	n/a

Table 3: Exact BPS spectra at $p = q = 3$, resolved into $U(N)$ irreps. The window is the same set of degrees in every irrep computed, of width $N + 1$ and centred on $pN^2/2$. “Index-invisible” is the fraction of BPS states not accounted for by the refined Witten index, which is a lower bound on the count and equals it only when the strand is concentrated.

2.6 Cohomology irrep by irrep

The bound is one thing and the spectrum is another. To get the latter we compute exactly. The complex splits over weights because Q is a gauge singlet, which replaces the 2^{pN^2} -dimensional Fock space by weight spaces of dimension at most a few times 10^4 ; within each weight space the ranks of the integer matrices Q_k are computed by Gaussian elimination over $\mathbb{F}_P$, with every rank confirmed against a second prime, and per-irrep multiplicities follow by Kostant inversion of $\dim H^k(W_\lambda) = \sum_{\mu \geq \lambda} K(\mu, \lambda) h_\mu^k$ from the top down. Two devices extend the reach: a right-looking blocked LU whose trailing update is a single BLAS matrix multiply, kept exact by choosing the prime small enough that float64 products are exact integers; and $\mathbb{Z}_p$ flavour blocking over a field containing a p -th root of unity, which splits each weight space into p blocks and buys a factor of p^2 in both time and memory while resolving the flavour charge for free.

Table 3 collects the results. Three features matter. First, the window is the same in every irrep computed, not merely at the top: at $N = 3$ all thirteen irreps down to $C_2 = 24$ have support exactly $\{12, 13, 14, 15\}$, including $(4, 0, -4)$ whose Fock multiplicity is 110 080, some 215 times that of the maximal irrep. Descending the Casimir ladder is what makes this a statement about the spectrum rather than about its thin top. Second, the index is a lower bound and not a proxy: the excess over the index is 10% at $N = 3$ and 19% at $N = 4$, so saturation, which is the sharp form of concentration, fails wherever the index is non-zero. Third, the profile $3^N \binom{N}{j}$ of Table 1 is not a law. It holds for eleven of the thirteen irreps at $N = 4$ and breaks at $C_2 = 107$, where the profile is 114, 1572, 2916, 1572, 114, with the centre unchanged and the window edges suppressed. The width is what is robust.

3 Density of states and the bootstrap

Nothing in Section 2 says anything about energies above zero. That is the gap the bootstrap fills.

3.1 The sector-resolved semidefinite programme

The object of interest is the ground energy $E_0(k)$ of the sector with $N_\Psi = k$. Rather than optimising over states we optimise over linear functionals $\phi(O) = \text{Tr}[\rho_k P_k O P_k]$, where ρ_k is Hermitian on the k -block and is *not* assumed positive:

$$\begin{aligned} \min_{\rho_k = \rho_k^\dagger} \phi(H) \quad \text{subject to} \\ \phi(\mathbf{1}) = 1, \quad [\phi(\text{Tr} w_a^\dagger w_b)]_{ab} \succeq 0, \quad [\phi((\text{Tr} w_a)^\dagger \text{Tr} w_b)]_{ab} \succeq 0, \quad \phi([H, X]) = 0, \quad \phi((N_\Psi - k)X) = 0. \end{aligned} \tag{13}$$

Here w runs over open, adjoint-valued words in the letters $\Psi^a, \bar{\Psi}^a$ up to a length L called the level. The two matrix inequalities are positivity of the Gram matrix of the states $w_{ij}|\psi\rangle$ in its adjoint and singlet channels; the equalities are the equations of motion and the statement that the functional lives in sector k , imposed for every operator X appearing in the cones. Every constraint is satisfied by the true sector ground state, so the minimum is a rigorous lower bound on $E_0(k)$ that can only improve with L , and the dual produces a sum-of-squares certificate $H - E \succeq 0$ on the sector.

Two ingredients matter most. The sector projection itself, since charged operators have $\phi = 0$ and commutators act within the block; and symmetry, because ρ_k may be taken invariant under $SU(N)_{\text{gauge}} \times \mathbb{Z}_3$ at no cost by group averaging, after which the singlet and adjoint channels separate as $\phi(\text{Tr} w_a^\dagger w_b) - \frac{1}{N} \phi(\text{Tr} w_a^\dagger \text{Tr} w_b) \succeq 0$ [5]. Implementing the invariance as an isotypic decomposition $\rho_k = \oplus_R \sigma_R \otimes \mathbf{1}$ leaves $D = \sum_R m_R^2$ parameters instead of d_k^2 , at most 90 at $N = 2, k = 2$ and 666 at $k = 3$.

sector k	0	1	2	3	4	5	6	7
exact $E_0(k)$ (full ED, 2^{12} states)	98	22	5.16536	1.22706	0.08796	0	0	0
best bootstrap lower bound	98	22	5.16536	1.22706	0	0	0	0
level at which the bound is exact	2	2	3	4	> 4	2	2	2

Table 4: Three-flavour model at $N = 2$. Sectors $k \geq 8$ follow by particle-hole symmetry $k \rightarrow 12 - k$. Every gapped sector except the almost-BPS $k = 4$ is reproduced exactly, and the window $\{5, 6, 7\}$ is exact automatically through the supersymmetric floor. A lifted sector becomes tight one level after it leaves the floor, and the singlet/adjoint split is what closes the gap: with the plain adjoint channel the $k = 2$ bound stalls at 4.55.

	$N = 3$	$N = 4$	$N = 5$	$N = 10$	$N = 100$
$E_0(2; N)$, bootstrap (level 2)	159.000	660.000	1545.000	15090.000	1.59909×10^7
$E_0(2; N)$, few-body ED	159	660	1545	n/a	n/a
$E_0(3; N)$, bootstrap (level 2)	45 (level 3: 70.35)	508.000	1355.000	14710.000	n/a
$E_0(3; N)$, few-body ED	75.7917	508	n/a	n/a	n/a

Table 5: Low sectors at larger N : rigorous lower bounds from one constraint set evaluated at each N , next to few-body exact diagonalisation where $\binom{3N^2}{k}$ permits it (n/a: out of reach). All tight entries equal (15). The one non-tight entry is the first interacting sector $k = N = 3$.

3.2 An exact finite- N formulation in trace words

The cost of (13) is still set by the block dimension, so to go beyond $N = 2$ we re-express the same constraints as identities among expectation values of trace words $\phi(\text{Tr} w_1 \cdots \text{Tr} w_m)$, with N entering only as a numerical coefficient. Only one algebraic fact is needed, that moving a letter cyclically around a trace produces double traces,

$$\text{Tr}[\ell_1 \ell_2 \cdots \ell_L] = (-1)^{L-1} \text{Tr}[\ell_2 \cdots \ell_L \ell_1] + \sum_m (-1)^m \delta_{\ell_1 \bar{\ell}_m} \text{Tr}[\ell_2 \cdots \ell_{m-1}] \text{Tr}[\ell_{m+1} \cdots \ell_L], \quad (14)$$

whereas swapping two adjacent letters inside a trace does not preserve the matrix structure, so no normal ordering is ever performed. The important point is what is not done: no large- N factorisation is imposed. Multi-trace expectation values are kept as independent unknowns, so every bound is exact at the chosen integer N , and the planar limit, which would wash out the $1/N$ -suppressed concentration physics, is never taken. The resulting programme is built once and evaluated at any N , with size set by the level rather than by N or k .

3.3 The free sectors in closed form

A single constraint set evaluated at $N = 2, 3, 4, 5, 10, 30, 100$ reproduces the analytic anchors $E_0(0) = 16N^3 - 15N$ and $E_0(1) = 16N^3 - 53N$ to eight digits. Beyond those, the level-2 bound is tight for $k = 2$ at every $N \geq 3$ and for $k = 3$ at every $N \geq 4$, and in all such cases equals the same expression; since a simple variational state gives the same energy from above, this establishes

$$E_0(k; N) = 16N^3 - (15 + 38k)N \quad \text{for } k \leq N - 1. \quad (15)$$

In words, the k -particle sector is free as long as $k \leq N - 1$. The lowest one-particle multiplet is an $SU(N)$ adjoint, and k antisymmetrised copies of it, in the component of $\wedge^k \mathbf{adj}$ of largest Casimir, are annihilated by the quartic interaction. We checked this explicitly at $N = 3$, where $\wedge^2 \mathbf{8} = \mathbf{10} \oplus \overline{\mathbf{10}} \oplus \mathbf{8}$ and the 20-dimensional part is free while the adjoint is lifted by +100. The first interacting sector is therefore $k = N$, and there the bootstrap gives the first rigorous bound beyond $N = 2$: $E_0(3; 3) \geq 70.35$ at level 3, against the exact 75.79 from few-body exact diagonalisation in the 2925-dimensional sector, level 2 returning only the free value 45.

3.4 Results

Tables 4 and 5 give the sector energies. The bootstrap also produces allowed regions rather than single numbers. Fixing $\phi(H) = E$ and extremising an observable maps out an island in the $(E, \langle \hat{C}_2 \rangle)$ plane enclosing the exact spectrum (Fig. 5); for the single-matrix model the island degenerates to the exact Casimir line, which is the cleanest possible illustration of what a Casimir Hamiltonian does to the constraint span. With eigenstate constraints $\phi(OH) = E \phi(O)$ added, the level-2 problem becomes feasible only at the exact eigenvalues, 5.16536, 12 and 13.4907, each to 10^{-8} : an archipelago rather than an island.

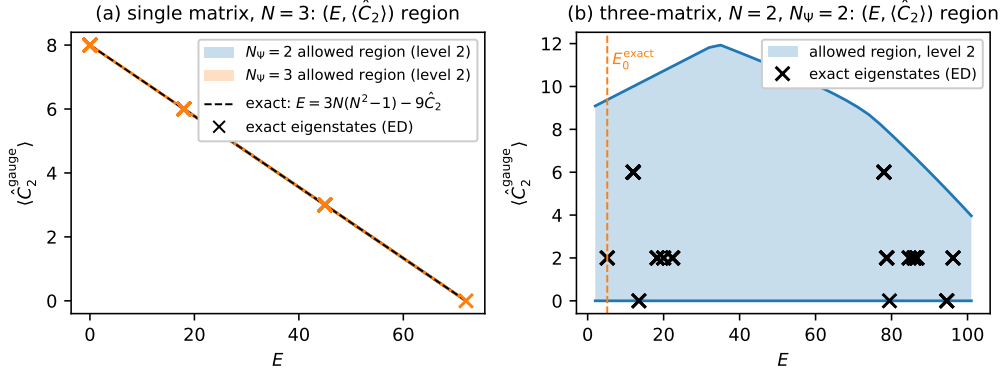


Figure 5: Allowed regions in the energy and gauge-Casimir plane from the level-2 sector bootstrap, obtained by fixing $\phi(H) = E$ and extremising $\phi(\hat{C}_2)$, against the exact eigenstates. (a) Single-matrix model at $N = 3$: the region has zero width and coincides with the exact line $E = 3N(N^2 - 1) - 9\hat{C}_2$, because that operator identity lies in the span of the constraints. (b) Three-flavour model at $N = 2$: a genuine two-dimensional island, rigorous and already tight at the ground-state edge.

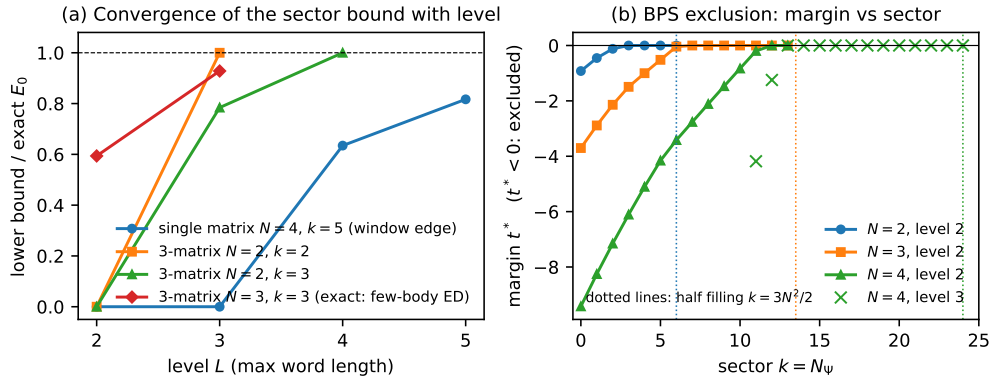


Figure 6: (a) Sector lower bound divided by the exact ground energy as a function of level, for four representative sectors. (b) BPS-exclusion margin as a function of the sector: a negative margin certifies that the sector contains no BPS state at that level.

BPS exclusion, and where the two methods meet. A BPS state satisfies $Q\psi = \bar{Q}\psi = 0$ and therefore $\phi(XQ) = \phi(QX) = \phi(X\bar{Q}) = \phi(\bar{Q}X) = 0$ for every X . Adding those rows to (13) and maximising a feasibility margin turns the bootstrap into a certificate of absence: a negative margin means the sector contains no BPS state. At $N = 2$ this excludes $k \leq 2$ at level 2 and $k \leq 3$ at level 3, one level earlier than the energy bound each time; at $N = 3$ and $N = 4$ level 2 already excludes $k \leq 6$ and $k \leq 11$ in seconds (Fig. 6b).

The test sharpens considerably once it is resolved by irrep, which is possible because the gauge Casimir is itself a trace polynomial (Section 4), so $\phi((\hat{C}_2 - c_\lambda)X) = 0$ is a family of linear rows of exactly the same type as the sector rows. Table 6 shows what happens in the maximal-Casimir irrep, where the largest complexes live.

This is the point of running both methods. The cohomology knows where the window is but says nothing about the sectors outside it beyond the fact that they are empty; the bootstrap knows nothing about the window but certifies emptiness sector by sector, and its certificate degrades precisely at the boundary the cohomology identifies. Neither result depends on the other, and the exclusions never contradict the exact counts, which is a genuine consistency check on two very different pieces of machinery. It also indicates the limit of the method as currently implemented: the margins shrink by an order of magnitude per sector as the window is approached, so reaching the edge from outside requires either higher level or sharper irrep resolution, for instance through the cubic Casimir, which is also a trace polynomial.

An analytic complement. Because Q raises N_Ψ by three, the sectors organise into three strands $\mathcal{H}_c \rightarrow \mathcal{H}_{c+3} \rightarrow \dots$, each split further by irrep and flavour charge, and the Euler characteristic of a strand is purely

$N = 3, C_2 = 48$: sector k	8	9	10	11	12	13
exclusion margin	-0.876	-0.340	-0.050	-0.003	-0.0000	-0.0000
BPS states from exact cohomology	none	none	none	none	27	81

Table 6: The bootstrap meeting the cohomology. In the maximal-Casimir irrep at $N = 3$ the BPS-exclusion margin is clearly negative, hence a certificate of absence, at $k = 8, 9, 10, 11$; it then decays to solver tolerance exactly at $k = 12$, which is where the exact computation of Section 2.6 places the lower edge of the window. The two calculations share no input. At $N = 4$ the same test in the $C_2 = 120$ irrep excludes $k = 18, 19, 20$ and loses resolution from $k = 21$, one sector short of the true edge at $k = 22$.

kinematic. Its modulus bounds the number of BPS multiplets below, with equality exactly when the strand is concentrated. Using $\mathbb{Z}_3$ -Fourier flavour letters the Fock character $\prod_m \prod_{i,j} (1 + tz^m u_i / u_j)$ gives every such index exactly, and the totals per strand have the closed form

$$I_c(N) = \frac{2}{3} 3^{3N^2/2} \cos\left(\frac{2\pi c}{3} + \frac{\pi N^2}{2}\right), \quad (16)$$

which is the Turiaci–Witten profile $\cos(\pi(k - k_*)/3)$ about half filling: 2:1:1 for even N and 0:1:1 for odd N . The per-irrep multiplicities grow like $e^{0.75N^2}$, being 18, 891 and 3.5×10^5 at $N = 2, 3, 4$, and include gauge singlets. The single-matrix model put through the same computation has a single irrep per strand, multiplicity at most 9, and no singlets at all.

4 Single-trace formulas for H , Q and $\hat{C}_2$

4.1 The Hamiltonian

Both the trace bootstrap and the Casimir-resolved exclusion need H and $\hat{C}_2$ as explicit trace polynomials. Writing $Q = C_{abc} \text{Tr}[\Psi^a \Psi^b \Psi^c]$ with C cyclic, defining the flavour tensors

$$K_{bc;ed} = \sum_a C_{abc} \bar{C}_{dea}, \quad M_{cd} = \sum_{ab} C_{abc} \bar{C}_{abd}, \quad L_{cd} = \sum_{ab} C_{abc} \bar{C}_{bad}, \quad \|C\|^2 = \sum |C_{abc}|^2, \quad (17)$$

and the matrix-valued composites $X_a = C_{abc} \Psi^b \Psi^c$, a Wick expansion of the anticommutator gives the normal-ordered canonical form

$$H = \{Q, \bar{Q}\} = 9 \sum_a \text{Tr}[X_a \bar{X}_a] - 9N M_{cd} \text{Tr}[\Psi^c \bar{\Psi}^d] + 9 L_{cd} \text{Tr} \Psi^c \text{Tr} \bar{\Psi}^d + 3N^3 \|C\|^2 - 3N \langle C, C^{\text{rev}} \rangle. \quad (18)$$

The structure is a single-trace quartic, a single-trace quadratic, a product of two length-one traces, and a constant. The quartic is the normal-ordered $|\partial Q|^2$, as it must be for $\{Q, \bar{Q}\}$ with Q built from creation operators only. The only genuine double trace is $\text{Tr} \Psi^c \text{Tr} \bar{\Psi}^d$, a bilinear in the p trace-mode fermions, of size $O(N)$ against the $O(N^3)$ of the other terms and therefore suppressed by $1/N^2$, though we keep it exactly at finite N . This was verified as an operator identity on random vectors to relative residual 8×10^{-16} at $(N, p) = (2, 1), (3, 1), (2, 2), (3, 2), (2, 3)$.

Specialising to the three-flavour model, $\|C\|^2 = 16/3$, $\langle C, C^{\text{rev}} \rangle = 5$, $M = \frac{5}{9} \mathbb{K} + \frac{11}{9} \mathbb{J}$ and $L = \frac{4}{9} \mathbb{K} + \frac{11}{9} \mathbb{J}$, so

$$H = 9 \sum_{a=1}^3 \text{Tr}[X_a \bar{X}_a] - 5N N_\Psi - 11N \sum_{c,d} \text{Tr}[\Psi^c \bar{\Psi}^d] + 4 \sum_c \text{Tr} \Psi^c \text{Tr} \bar{\Psi}^c + 11 \sum_{c,d} \text{Tr} \Psi^c \text{Tr} \bar{\Psi}^d + 16N^3 - 15N. \quad (19)$$

Reading off the vacuum energy gives $E_0(0) = E_0(3N^2) = 16N^3 - 15N$, which is 98 at $N = 2$ and 387 at $N = 3$, and the one-particle sector gives gauge-traceless states at $16N^3 - 53N$ and $16N^3 - 20N$ and trace-mode states at $16N^3 - 16N$. These are the analytic anchors used to validate the bootstrap in Section 3. Note also that the trace modes do not decouple at $p = 3$ as they do at $p = 1$: expanding Ψ^a into trace and traceless parts, Q retains a term proportional to $C_{a[bc]}$, which is non-zero for the three-flavour couplings, so the $\mathfrak{su}(N)$ version of the model is a genuinely different theory.

4.2 The gauge Casimir

Under $\Psi^a \rightarrow U \Psi^a U^\dagger$ the one-body generator for traceless T is $E(T) = \sum_{ii'} T_{ii'} M_{ii'}$ with the matrix word $M = \sum_a (\Psi^a \bar{\Psi}^a + \bar{\Psi}^a \Psi^a)$, and the completeness relation for $su(N)$ gives

$$\hat{C}_2 = \sum_x J^x J^x = \frac{1}{2} \text{Tr}[M^2] - \frac{(\text{Tr} M)^2}{2N} = \frac{1}{2} \sum_{a,b} \text{Tr}[(\Psi^a \bar{\Psi}^a + \bar{\Psi}^a \Psi^a)(\Psi^b \bar{\Psi}^b + \bar{\Psi}^b \Psi^b)] - \frac{1}{2} p^2 N^3, \quad (20)$$

a trace polynomial of degree four, flavour invariant, with eigenvalue $c_\lambda = \frac{1}{2} \sum_i \lambda_i (\lambda_i + N + 1 - 2i)$ on the irrep λ . This was verified against the explicit $\sum_x J^x J^x$ to 10^{-16} at $(N, p) = (2, 3), (2, 1), (3, 1)$. Equation (20) is what makes the irrep-resolved exclusion of Table 6 possible: a functional supported on the $\hat{C}_2 = c_\lambda$ eigenspace obeys $\phi((\hat{C}_2 - c_\lambda)X) = 0$, linear rows of the same form as the sector rows.

4.3 Why the model is not Casimir-solvable

For a single flavour, $H = 3N(N^2 - 1) - 9\hat{C}_2$ and everything is determined by representation theory. That mechanism is gone at $p = 3$, and in two independent ways. Structurally, the quartic part $9K_{bc;ed} V^{bced}$ contains no alternating words $\text{Tr}[\Psi \bar{\Psi} \Psi \bar{\Psi}]$, while (20) does, and K is not proportional to the Casimir flavour structure: projecting it gives the component $-\frac{16}{3}\hat{C}_2$ against $-9\hat{C}_2$ at $p = 1$, but the remainder is non-zero. Numerically at $N = 2$, resolving N_Ψ , $\hat{C}_2$, J_z and the $\mathbb{Z}_3$ charge simultaneously splits the 4096-dimensional space into 353 joint sectors, and in 245 of them the restriction of H has more than one distinct eigenvalue, which a function of Casimirs cannot. The same computation shows the commutant of H among one-body operators is exactly four-dimensional, $\text{span}(J^1, J^2, J^3, N_\Psi)$, so there is no hidden continuous symmetry and in particular no conserved flavour charge.

This is the reason the bootstrap of Section 3 has to be sector-resolved. When H is a Casimir function the operator identity lies in the span of the constraints, the allowed region collapses to a line (Fig. 5a), and nothing has to be solved. When it is not, $E_0(N_\Psi, \lambda)$ is a genuine dynamical quantity, the supersymmetric floor $\phi(H) \geq 0$ has to be lifted by constraints that actually see the sector, and the island acquires two dimensions (Fig. 5b). It is also the reason $p = 3$ was singled out in the first place: it is the smallest flavour number for which the Casimir shortcut fails, and our own cohomology confirms the same transition from a different direction, since at $p = 1$ the BPS states occupy a single irrep while at $p \geq 2$ they are spread over every irrep the Fock space contains.

5 Status

The exact side is in good shape at the top of the spectrum and thins out below it. The rank law (6) and the trichotomy (9) are derived; the lower bound (12) is derived and tight for $p \leq 3$; the irrep-resolved spectra of Table 3 are exact but reach only thirteen irreps at $N = 3$ and $N = 4$, and the complexes with genuinely macroscopic index, which is what the concentration conjecture is about, remain out of reach without a sparse exact rank routine. Two statements are verified rather than proved: that the window is the same in every irrep, and that a generic q -form saturates its index class by class.

The bootstrap side is rigorous wherever it returns a number, but its reach is set by the level. It is exact in every gapped sector at $N = 2$, gives (15) for the free sectors at any N , and provides one rigorous bound in an interacting sector beyond $N = 2$. The natural next steps are the level-4 problem, which is assembled but not yet solved to tolerance, and near-BPS energies as a function of N in the first interacting sector $k = N$, which is the direct test of super-Schwarzian scaling and the one question neither method has yet answered.

A Numerical implementation of the bootstrap

Finite- N engine. Open words are built as blocks between sectors, so the 2^{pN^2} -dimensional space is never formed. An isotypic reducer maps every operator to its $D = \sum_R m_R^2$ invariant components as soon as it is built, and the spans of the touched operators and of the equations of motion are accumulated as streaming $2D \times 2D$ Gram matrices. A diagnostic that evaluates every constraint on the exact, multiplet-averaged ground state caught a round-off Hermitian part being normalised into a spurious constraint; all such cutoffs are now relative to $\|X\|$.

Trace algebra. One primitive generates everything: swapping two adjacent letters in a routed term, meaning letters in operator order together with a routing permutation, where the anticommutator branch rewires the

routing and closed index loops give factors of N . From it follow canonical cyclic forms, products, daggers, commutators with H , self-symmetry reductions such as $\text{Tr}[\ell\ell] = 0$, and the finite- N relations from the vanishing antisymmetriser over $N + 1$ indices. Coefficients are polynomials in N . Every identity is checked against explicit operators at $N = 2, 3$, a thousand random terms to 10^{-15} , and the algebra recovers $\{Q, \bar{Q}\} = H$ on its own.

Assembly and conditioning. Variables are generated by closure; reality is imposed at the variable level, halving the unknowns; rows are unit-normalised; the unknowns carry a 't Hooft scaling $N^{\sum(1+L_i/2)}$ and the objective is divided by N^3 , without which nothing works beyond $N \approx 30$. Solvers are called directly. The settings that matter are Clarabel with chordal decomposition off and equilibration on, and SCS with $\rho_x = 10^{-3}$, the Apple-Accelerate LDL backend and fixed scaling. A parallel fixed- N build produces the level-4 problem, 51k monomials with cones up to dimension 168, in 25 seconds.

Exact cohomology. For Section 2.6, ranks are computed over $\mathbb{F}_P$ with $P = 2^{31} - 1$ and cross-checked against $P' = 2147483629$. The blocked routine uses a prime small enough that $b(P - 1)^2 < 2^{53}$, so that a float64 matrix multiply of inner dimension b is exact integer arithmetic and can be reduced afterwards; this makes the trailing update of the elimination a single BLAS call. Flavour blocking over a field containing a p -th root of unity splits each weight space into p blocks. The largest weight space handled is 30 294 after blocking, in about thirty minutes at 6.1 GB.

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